

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: Database Management Systems
COURSE CODE: CSA05

1

```
SELECT c.CustomerID, c.CustomerName, o.OrderID
FROM Customer c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

2

```
SELECT EmployeeName
FROM Employee e
WHERE Salary >
(
    SELECT AVG(Salary)
    FROM Employee
    WHERE DepartmentID = e.DepartmentID
);
```

3

```
CREATE VIEW EmpView AS
SELECT DepartmentID,
    COUNT(*) AS TotalEmployees,
    AVG(Salary) AS AvgSalary
FROM Employee
GROUP BY DepartmentID;
```

4

```
Employee ⋈ Employee.DepartmentID = Department.DepartmentID Department
```

5

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    FOREIGN KEY (CustomerID)  
    REFERENCES Customer(CustomerID)  
);  
  
INSERT INTO Customer(CustomerID, CustomerName)  
VALUES (500, 'John');  
  
INSERT INTO Orders(OrderID, CustomerID)  
VALUES (101, 500);
```

6

```
SELECT e.EmployeeName  
FROM Employee e  
JOIN Department d  
ON e.DepartmentID = d.DepartmentID  
WHERE d.Location = 'Chennai';
```

7

```
SELECT DepartmentID,  
    COUNT(*) AS TotalEmployees  
FROM Employee  
GROUP BY DepartmentID;
```

8

```
UPDATE Employee  
SET Salary = Salary + 5000  
WHERE EmployeeID = 101;
```

9

```
SELECT e.EmployeeName
FROM Employee e
JOIN Department d
ON e.DepartmentID = d.DepartmentID
JOIN Manager m
ON d.ManagerID = m.ManagerID
WHERE m.Experience > 10;
```

10

$$\{E \mid \exists D (Department(D) \wedge E.DepartmentID = D.DepartmentID)\}$$